Anurag Karki

513-400-2285 | karkiag@mail.uc.edu | Linkedin | Homepage

Summary

Mechanical Engineer specializing in UAV design, simulation, and automation. Experienced in integrating hardware and software for autonomous flight systems, with strong proficiency in Python, MATLAB, and ANSYS. Passionate about advancing reliable, high-performance aerial platforms through data-driven design and testing.

Education

University of Cincinnati — Cincinnati,OH
Master of Science in Mechanical Engineering — *Aug. 2024 – Apr. 2026*

- 3.867/4.0 GPA
- Courses Undertaken: Design of Experiment, Decision Engineering, Industrial AI, Intelligent System , Introduction to Robotics, Robot Control and design

Tribhuvan University — Kathmandu, Nepal
Bachelor in Mechanical Engineering — *Nov. 2017 – Jul. 2022*

- 3.7/4.0 GPA
- Electives taken: Advance Mechanical Design, Mechanical Design & Simulation, Operation Research

Experience

Graduate Research Assistant — June 2025 – Present
P&G Digital Accelerator @ University of Cincinnati — *Cincinnati, OH*

- Executed Design of Experiments (DOE) to analyze paper property variations using ABAQUS simulation.
- Developed Python scripts to automate and streamline ABAQUS simulation workflows
- Designed and implemented VBA-based Excel tools

Mechanical Engineer — Nov 2022 - Jan 2024
National Innovation Center Nepal — *Kathmandu, Nepal*

- Designing 3D CAD models and assemblies of fixed-wing UAVs using SolidWorks
- Performing Finite Element Analysis (FEA) using ANSYS to evaluate structural integrity of UAV frames
- Collaborating with electronics and AI teams to integrate sensors, avionics, and battery modules into airframes
- Leading end-to-end UAV prototyping using 3D printing, laser cutting, and carbon-fiber layups for functional testing

Mechanical Engineer — Jul 2022 - Nov 2022
Incubation, Innovation, and Entrepreneurship Center (IIEC) — *Lalitpur, Nepal*

- Designing modular UAV components for rapid prototyping and iteration
- Assisting with ROS-based test flights for navigation stack validation in simulated and physical environments
- Modeling UAV frame structures and sensor payload mounts in SolidWorks for use in Software-in-the-Loop (SITL) and Hardware-in-the-Loop (HITL) environments

Projects

Aerodynamic Shape Optimization of Blended Wing Body Vehicle — Jan 2021 - Feb 2022

- Conducted both aerodynamic and stability analysis of a blended-wing body (BWB) vehicle
- Optimized the shape of the planform with 23% increase in aerodynamic efficiency

Design and fabrication of Long endurance Unmanned aerial vehicle (UAV) — Jul 2020 - Nov 2020

- Implemented iterative design process to come to a final design selection
- CAD modeled the entire vehicle
- Fabricated parts using 3D-printer, laser-cutter

Design and Testing of Decision Support Mechanism — Jan 2021 - Feb 2022

- Designed a data acquisition system using Arduino and sensors

- DAS was attached to UAV and communicated using telemetry to ground station
- Decision support mechanism was built using DAS

Genetic Algorithm-based UAV Path Planning for Wildlife Monitoring Oct 2024 - Mar 2025

- Developed an automated path planning system for UAVs using a Genetic Algorithm approach
- Optimized flight paths to maximize the probability of wildlife detection
- Achieved improved performance compared to existing path planning methods

Technical Skills

Design Skills: Solidworks, XFLR5, OpenVSP
Programming Skills: Python , MATLAB, ROS2
Computational Skills: ANSYS, SU2, SIMULINK
Miscellaneous Skills: Linux, Latex, MS Office